

AI & ML Lab - Final Lab Assignment Links

Name: Ahsanul Haque

ID: 220119

CNN Image Classification - Geometric Shapes

Repository and Notebooks

1. **GitHub Profile:**
<https://github.com/ahsanjust/>
2. **Final Lab Standalone Assignment Repository:**
<https://github.com/ahsanjust/CNN-Geometric-Shapes-Classification>
3. **CNN - GitHub Notebook:**
https://github.com/ahsanjust/CNN-Geometric-Shapes-Classification/blob/master/220119_CNN.ipynb
4. **Google Colab Notebook (GitHub-Native, Direct Link):**
https://colab.research.google.com/github/ahsanjust/CNN-Geometric-Shapes-Classification/blob/master/220119_CNN.ipynb
5. **Repository README:**
<https://github.com/ahsanjust/CNN-Geometric-Shapes-Classification/blob/master/readme.md>

Performance Summary - CNN (Standard Test Set)

Class	Precision	Recall	F1-Score	Support
Circles	1.0000	0.9000	0.9474	30
Squares	1.0000	1.0000	1.0000	30
Triangles	0.9091	1.0000	0.9524	30
Overall Accuracy	0.9556 (95.56%)			

Architecture & Workflow Summary

- **Model:** 2-Stage PyTorch CNN
Conv2d(3→32) → ReLU → MaxPool(2) →
Conv2d(32→64) → ReLU → MaxPool(2) →
Flatten → Linear(16384→128) → ReLU → Linear(128→3)
- **Input Preprocessing:** 64 × 64 RGB images with single shared transform:
Resize → ToTensor → Normalize($\mu = 0.5, \sigma = 0.5$)
- **Training Setup:** CrossEntropyLoss, Adam optimizer (lr = 0.001), batch size 64, 10 epochs (Parameters: 2,117,059)

- **Dataset:** Geometric Shapes (Option 5) - train 543, validation 180, test 90, plus 15 custom smartphone photos
- **Real-World Accuracy:** 12 / 15 (80.0%) on custom smartphone photos
(Circles: 4/5, Squares: 3/5, Triangles: 5/5)
circle4.jpg correctly classified at 98.9% confidence
- **Zero-Manual-Upload Automation:** Clones repo automatically in Google Colab, loads weights from model/220119.pth, evaluates test set, and predicts on custom photos automatically upon clicking "Run All".